

Suozhi Huang

Ph.D. Student in Electrical and Computer Engineering, Princeton University

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RESEARCH PROFILE

Ph.D. student researching LLM reasoning, post-training, and agentic systems for reliable decision-making. Co-first author of *InternLM2.5-StepProver* (ICML 2025), *LeanProgress* (TMLR 2025), and *CryptoBench*; coauthor of the NeurIPS 2025 Best Paper. Google Scholar: 577 citations, h-index 6 (September 2026).

EDUCATION

Princeton University

Ph.D. Student, Electrical and Computer Engineering; Advisor: Prof. Mengdi Wang

Princeton, NJ

Sep 2025 – Present

Tsinghua University

B.E., Computer Science and Technology (Artificial Intelligence), Yao Class

Beijing, China

Sep 2020 – Jun 2025

RESEARCH EXPERIENCE

Princeton University

Graduate Researcher; Advisor: Prof. Mengdi Wang

Princeton, NJ

Sep 2025 – Present

- Co-developed **CryptoBench**, a dynamic benchmark with 50 monthly expert-curated tasks spanning on-chain analysis, retrieval, and prediction; evaluated 10 leading LLMs in direct and agentic settings.
- Contributed to **AQuA**, two recursively self-improving research-agent systems for quantitative investment; the systems achieved a combined information coefficient of 0.190 on crypto and a held-out Sharpe ratio up to 2.50 on U.S. equities after two-leg costs.

Shanghai AI Laboratory

Research Intern, Formal Mathematical Reasoning; Advisors: Dr. Zheng Yuan and Prof. Dahua Lin

Shanghai, China

Jun 2024 – Oct 2024

- Co-developed critic-guided proof search and expert iteration for **InternLM2.5-StepProver**, improving the miniF2F test pass rate from 54.5% to 65.9%.
- Built the critic-model training and data pipeline, including preference pairs from Lean proof-search trajectories; helped scale expert iteration to 2.19B tokens and more than 20,000 CPU-days.

Anima AI+Science Lab, Caltech

Researcher, Neural Theorem Proving; Advisor: Prof. Anima Anandkumar

Pasadena, CA

Aug 2024 – Jan 2025

- Co-led and developed **LeanProgress** as a co-first author, training a proof-progress predictor on approximately 80K Lean trajectories; achieved 75.8% exact-match accuracy for remaining-step prediction.
- Integrated progress signals into ReProver search, raising the Mathlib4 test pass rate from 41.4% to 45.2%; contributed tooling merged into LeanDojo v2.

New York University

Visiting Researcher, Collaborative Perception; Advisors: Prof. Chen Feng and Dr. Yiming Li

New York, NY

Feb 2023 – Sep 2023

- Co-developed **ActFormer**, a pose-guided Transformer for scalable multi-robot perception that improved AP@0.7 from 29.89% to 45.15% while using approximately 50% fewer BEV queries.
- Adapted V2X-Sim to MMDetection3D and led multi-agent camera experiments and ablations; published as a co-first-author paper at ICRA 2024.

SELECTED PUBLICATIONS

AQuA: Recursively Self-Improving Quantitative Trading Research Agents

Jiacheng Guo, **Suozhi Huang**, Yunlong Gao, Zihao Li, Jason Ge, Xu Kuang, Mengdi Wang

arXiv:2608.12841, 2026. [Paper](#)

Gated Attention for Large Language Models: Non-linearity, Sparsity, and Attention-Sink-Free

Zihan Qiu, Zekun Wang, Bo Zheng, et al., **Suozhi Huang**, et al.

NeurIPS 2025, Best Paper Award. [Paper](#)

InternLM2.5-StepProver: Advancing Automated Theorem Proving via Critic-Guided Search

Zijian Wu*, **Suozhi Huang***, Zhejian Zhou, Huaiyuan Ying, Zheng Yuan, Wenwei Zhang, Dahua Lin, Kai Chen

ICML 2025. [Paper](#) | [Code](#)

LeanProgress: Guiding Search for Neural Theorem Proving via Proof Progress Prediction

Robert Joseph George*, **Suozhi Huang***, Peiyang Song, Anima Anandkumar

Transactions on Machine Learning Research, 2025. *Co-first author*. [Paper](#)

CryptoBench: A Dynamic Benchmark for Expert-Level Evaluation of LLM Agents in Cryptocurrency

Jiacheng Guo*, **Suozhi Huang***, Zixin Yao*, Yifan Zhang*, Yifu Lu*, et al.

arXiv:2512.00417, 2025. [Paper](#)

Any2Poster: Any-Source Poster Generation Across Modalities and Domains

Amogh Vinaykumar, Aiden Li, **Suozhi Huang**, Shilong Liu

arXiv:2606.02915, 2026. [Paper](#) | [Code](#)

FutureX-Pro: Extending Future Prediction to High-Value Vertical Domains

Jiashuo Liu, Siyuan Chen, Zaiyuan Wang, Zhiyuan Zeng, Jiacheng Guo, et al., **Suozhi Huang**, et al.

arXiv:2601.12259, 2026. [Paper](#)

ActFormer: Scalable Collaborative Perception via Active Queries

Suozhi Huang*, Juexiao Zhang*, Yiming Li, Chen Feng

IEEE International Conference on Robotics and Automation (ICRA), 2024. [Project](#)

A Knowledge–Data Dual-Driven Framework for Predicting Rechargeable Battery Electrolyte Properties

Yuchen Gao, Yu-Hang Yuan, **Suozhi Huang**, et al.

Angewandte Chemie International Edition, 2025. [Paper](#)

* *Equal contribution. Full publication list available on [Google Scholar](#).*

TECHNICAL SKILLS

Programming Python, C++, Lean 4, Coq

Machine Learning PyTorch, supervised fine-tuning, preference modeling, critic-guided search, LLM-agent evaluation

Systems and Tools Linux, Git, OpenCV, MMDetection3D, LeanDojo, ReProver

HONORS

NeurIPS 2025 Best Paper Award (coauthor)

2025

Gold Award, 36th Chinese Physics Olympiad (Top 30 nationally)

2019